MSAI 699 Capstone Project

Capstone Project Proposal:

NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence Platform for Personalized Cognitive Health Education Using Machine Learning and Retrieval-Augmented Generation

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Abstract

Innovation in AI technologies such as natural language processing, machine learning, and large language models have revolutionized access to information about the healthcare field (Bender et al., 2021; Lewis et al., 2020). Nevertheless, general-purpose AI systems produce misinformation because of the absence of mechanisms of validation of their answers. It is particularly problematic when it comes to cognitive health education, as patients and caregivers need factual information about Alzheimer's disease, mild cognitive impairment, genetics, and lifestyle interventions (Alzheimer's Association, 2024; World Health Organization, 2023).

NeuroGuide AI is the proposed solution – an explainable multi-agent artificial intelligence platform that utilizes Retrieval-Augmented Generation (RAG), machine learning, explainable artificial intelligence, and dedicated AI agents for delivery of personalized cognitive health education. This system will provide retrieval and validation of information about the medical sources, as well as risk assessments for educational purposes with help of machine learning and SHAP feature level explanations (Breiman, 2001; Lundberg & Lee, 2017).

Evaluating this project will be done through metrics of retrieval quality, response grounding, machine learning performance, explainability, and usability of the technology. The purpose of this research is to show that machine learning, RAG, XAI, and multi-agent architecture of the system can increase the accuracy and transparency of AI assisted cognitive health education.

Proposal

Project Title

NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence Platform for Personalized Cognitive Health Education Using Machine Learning and Retrieval-Augmented Generation

Problem Statement, Motivation, and Impact

Large language models have revolutionized the way individuals access healthcare-related information by offering a conversational medium for accessing difficult medical knowledge. Nonetheless, even with these advancements, most artificial intelligence systems are still vulnerable to issues such as hallucinations, outdated information, and lack of transparency. All these factors can lower the trust users have on such technology, thus affecting their decision-making concerning their healthcare needs (Bender et al., 2021). In cognitive health education, patients, their caretakers, and families require accurate information about Alzheimer's disease, mild cognitive impairments, genetic, and lifestyle-related information (Alzheimer's Association, 2024; World Health Organization, 2023).

This project aims at creating an innovative technology that would help deliver trustworthy cognitive health education using multi-agent artificial intelligence called NeuroGuide AI, with technologies such as Retrieval-Augmented Generation, machine learning, explainable AI, and other specialized AI agents.

The motivation behind this study is to prove that several approaches from the field of AI can be incorporated within a user-oriented platform aimed at health education. The improvements made to factual correctness, explainability, and personalization make NeuroGuide AI capable of boosting users' trust in health education using AI as well as highlighting the use of machine learning, natural language processing, explainable AI, and multi-agent systems.

Research Questions

Primary Research Question

Can an explainable multi-agent Retrieval-Augmented Generation (RAG) architecture provide more accurate, transparent, and trustworthy cognitive health education than a traditional single-agent conversational AI system?

Secondary Research Questions

RQ1: Does a multi-agent architecture improve the quality and reliability of educational responses compared to a single-agent approach?

RQ2: Can machine learning–based educational risk assessments improve personalization while maintaining transparency through explainable AI?

RQ3: How does Retrieval-Augmented Generation improve factual grounding and reduce unsupported responses?

RQ4: How do users perceive the usability, trustworthiness, and educational value of a multi-agent cognitive health platform?

Hypothesis

It is hypothesized that NeuroGuide AI will produce more accurate, transparent, and trustworthy educational responses than a traditional single-agent conversational AI system. By combining Retrieval-Augmented Generation, machine learning, explainable AI, and a multi-agent architecture, the platform is expected to improve factual grounding, reduce unsupported responses, and increase user confidence while providing personalized educational guidance.

Proposed AI Technologies

NeuroGuide AI integrates multiple artificial intelligence concepts explored throughout the Master of Science in Artificial Intelligence program.

Machine Learning

- Random Forest classification (Breiman, 2001)
- Educational risk prediction
- SHAP explainability

Natural Language Processing

- Large Language Models (GPT-4o-mini)
- TF-IDF document retrieval
- Latent Semantic Analysis (LSA)
- Prompt engineering

Retrieval-Augmented Generation

- Hybrid sparse and semantic document retrieval (Lewis et al., 2020)
- Citation-grounded response generation
- Confidence scoring

Multi-Agent Artificial Intelligence

The platform consists of five specialized AI agents (Wu et al., 2023). That independently perform retrieval, reasoning, validation, and response synthesis.

Human-Computer Interaction

- Personalized user interface
- Patient and clinician response modes
- Accessibility-focused interface design

Responsible AI

- Explainable AI (SHAP)

- Bias awareness
- Source transparency
- Educational medical disclaimers

Proposed System Architecture

The NeuroGuide artificial intelligence system analyzes all user requests via a set of five different AI agents working together in harmony. In contrast to one language model being used, each particular agent is responsible for a specific task contributing to the high accuracy, integrity, and reliability of the final output.

Model architecture:

User

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Personalization

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Retrieval Agent

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Reasoning Agent

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Validation Agent

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Synthesis Agent

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User

The proposed architecture consists of five collaborating AI agents.

Personalization

User profile information, including demographic and educational risk factors, is used to personalize educational responses and risk assessments.

Retrieval Agent

Searches trusted cognitive health resources and retrieves the most relevant supporting evidence.

Reasoning Agent

Analyzes retrieved information and generates evidence-based educational explanations tailored to the user's request.

Validation Agent

Verifies that generated responses are supported by retrieved evidence and assigns a confidence score to improve transparency.

Synthesis Agent

Produces the final educational response while adapting language for either patient or clinician audiences and providing supporting citations.

Datasets

The knowledge base will consist of trusted public sources related to Alzheimer's disease and cognitive health, including:

- PubMed Central
- NIH MedlinePlus (National Institute on Aging, 2024)
- Alzheimer's Association (Alzheimer's Association, 2024)
- Cochrane Reviews
- World Health Organization (WHO) (World Health Organization, 2023)
- Centers for Disease Control and Prevention (CDC)

The educational risk assessment model is based on publicly available dementia risk factors summarized by the **2024 Lancet Commission** (Livingston et al., 2024), incorporating demographic, lifestyle, cardiovascular, and APOE ε4 genetic information.

Evaluation

The effectiveness of NeuroGuide AI will be evaluated using quantitative and qualitative methods to assess system performance, machine learning accuracy, explainability, and user experience.

AI Performance

The Retrieval-Augmented Generation system will be evaluated using:

- Retrieval relevance
- Citation grounding

- Response confidence
- Response latency

Machine Learning

The educational risk assessment model will be evaluated using standard machine learning metrics, including:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Explainable AI

Model transparency will be assessed using SHAP feature attribution to verify that educational risk predictions are explainable and clinically reasonable (Lundberg & Lee, 2017).

Human-Computer Interaction

User experience will be evaluated through:

- System Usability Scale (SUS)
- User satisfaction
- Accessibility review
- Response readability

These evaluation methods will determine whether the proposed multi-agent architecture improves the reliability, transparency, and educational value of AI-assisted cognitive health education.

Ethical Considerations

NeuroGuide AI works within the field of healthcare education, which means that ethical AI use is crucial in this project (Bender et al., 2021). The system will operate only as an educational tool and won't give any diagnoses or suggestions on treatments. All answers will contain disclaimers urging people to seek assistance from licensed healthcare providers regarding any medical concerns.

For additional transparency, all answers will contain citations of reliable medical sources wherever possible. Explanatory approaches such as SHAP will help users comprehend how particular risk factors influence educational risk assessments.

Protected health information will not be collected and stored in the project. Possible sources of biases will be accounted for during the development of the project through using evidence-based medical resources and assessing the fairness of the answers.

Expected Challenges

There are some difficulties which are expected to occur while developing NeuroGuide AI. Firstly, it would be necessary to make sure that the response generated by the AI is accurate and comes from trusted medical sources. It will be solved using the techniques such as Retrieval-Augmented Generation, source citations and validation agent.

Another difficulty may be related to finding balance between personalization and transparency. Explainable AI using SHAP will allow users to better understand how each factor affects the result while emphasizing on the fact that the platform is purely an educational one. The last difficulty will be integration of several AI elements into user-friendly design.

Conclusion

The NeuroGuide AI system is an example of how a number of artificial intelligence approaches can be used together on one educational platform to generate reliable information related to cognitive health care. The use of machine learning, explainable AI, Retrieval-Augmented Generation, and a multi-agent system aims to increase the efficiency and personalization of AI-based health care education in addition to demonstrating knowledge gained during the MSAI program. (Breiman, 2001; Lewis et al., 2020; Lundberg & Lee, 2017).

AI Disclosure: Artificial intelligence was used only to assist with proofreading, grammar refinement, formatting, and organization. The project idea, research direction, system design, implementation, technical decisions, analysis, and all final content were developed independently by the author, who assumes full responsibility for this submission.

References

Alzheimer's Association. (2024). *2024 Alzheimer's disease facts and figures*. *Alzheimer's & Dementia, 20*(5), 3708–3821. https://doi.org/10.1002/alz.13809

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). *On the dangers of stochastic parrots: Can language models be too big?* In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 610–623).

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32.

Centers for Disease Control and Prevention. (2024). *Healthy aging and Alzheimer's disease*. https://www.cdc.gov/aging

LangChain. (n.d.). *LangGraph documentation*. https://langchain-ai.github.io/langgraph/

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). *Retrieval-augmented generation for knowledge-intensive NLP tasks*. *Advances in Neural Information Processing Systems, 33*, 9459–9474.

Livingston, G., et al. (2024). The 2024 Lancet Commission on dementia prevention, intervention, and care. *The Lancet, 404*(10452), 572–628.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*.

National Institute on Aging. (2024). *Alzheimer's disease fact sheet*. https://www.nia.nih.gov

World Health Organization. (2023). *Dementia*. https://www.who.int/news-room/fact-sheets/detail/dementia

Wu, Q., et al. (2023). *AutoGen: Enabling next-generation LLM applications via multi-agent conversation*. *arXiv*.